AICS-202 | CONSOLIDATED ADVERSARIAL MACHINE LEARNING ASSESSMENT DOSSIER

AICS-202 | Applied Adversarial Machine Learning & Defense

Repository Link:
https://github.com/YawukalMelaku/AICS-202-Adversarial-ML-Dossier

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EXECUTIVE SUMMARY

This assessment report evaluates the security posture and operational robustness of a multi-class image classifier trained on normalized $8\times88\times8$ pixel handwritten digits. Operating under a white-box threat model, the evaluation systematically measured system resilience across two primary attack vectors: inference-time input evasion using the Fast Gradient Sign Method (FGSM) and Projected Gradient Descent (PGD), and training-time data integrity compromise via label poisoning.

The baseline classifier achieved a clean-test accuracy of 96.22%. However, empirical testing demonstrated severe vulnerability to adversarial manipulation. Under an inference-time FGSM perturbation budget of $\epsilon=0.12\epsilon=0.12$, model test accuracy collapsed to 51.56%, and dropped further to 20.44% at $\epsilon=0.18\epsilon=0.18$. Multi-step iterative PGD attacks ($\epsilon=0.12,\text{steps}=7\epsilon=0.12,\text{steps}=7$) reduced accuracy to 48.89%.

A defensive retraining strategy utilizing naive FGSM sample augmentation was evaluated. Adaptive testing revealed that while clean accuracy dropped slightly to 94.00%, robustness under adaptive FGSM and PGD attacks further regressed to 28.44% and 26.00%, respectively. This outcome highlights the failure of simplistic adversarial data augmentation against adaptive opponents. Consequently, the candidate defense variant is REJECTED for production deployment pending advanced architectural hardening and input verification controls.

1. LABORATORY 01: BASELINE MODEL & THREAT MODELING

1.1 Baseline System Configuration

- Dataset Identifier: scikit-learn load_digits (1,797 total samples, 64 features per sample).

- Data Preprocessing: Grayscale pixel intensities normalized from $[0,16]$ to $[0.0,1.0]$.
- Data Partitioning: Stratified 75/25 train/test split (Seed: 42; Train $n=1,347$, Test $n=450$).
- Model Classifier: Multinomial Logistic Regression (max_iter=600, random_state=42).

Field	Assessment / Student Response
Protected Model & Asset	Local multi-class logistic regression classifier and its inference outsputs.
Legitimate Model Function	Accurately classifying authorized 8×8 grayscale handwritten numeral images ($0-9$).
Potential Attacker-Controlled Input	Pixel feature values of submitted inference digit images, normalized within $[0.0,1.0]$.
White-Box / Black-Box Assumption	White-box access (full model parameter knowledge and direct access to loss gradients).
Desired Adversarial Outcome	Untargeted or targeted misclassification of input digits via bounded perturbation.
Likely Monitoring / Controls	Input validation, $L\infty$ perturbation bounds, anomaly detection, data provenance, and human review.

- Clean Baseline Test Accuracy: 96.22% (433 correct out of 450 test samples).

1.2 Threat Model Worksheet

Field	Student Response / Operational Boundary
Protected Model & Asset	Locally trained multi-class logistic regression classifier and its prediction outputs
Legitimate Model Function	Correctly classifying authorized grayscale images of handwritten numerals
Potential Attacker-Controlled Input	Inference-time pixel intensity values (64 normalized input features) submitted to the model API/endpoint.
White-Box or Black-Box Assumption	White-box access (assuming full attacker visibility into model weights, coefficients, architecture, and direct access to input loss gradients).
Desired Adversarial Outcome	Untargeted misclassification of input digits via minimal, bounded perturbation
Likely Monitoring / Prevention Controls	Input validation, perturbation bounds, anomaly detection, adversarial retraining, distribution shift monitoring, and human review of critical predictions

1.3 Reflection & Discussion

Question: Why does good clean-test accuracy not establish security against manipulated inputs?

Response:
Clean-test accuracy evaluates model performance exclusively on unperturbed samples drawn from a benign, independently and identically distributed dataset. While high clean accuracy indicates solid generalization under standard operating conditions, it does not account for the presence of fragile high-dimensional decision boundaries. Adversarial manipulation involves an active adversary who explicitly calculates loss gradients to construct bounded perturbations designed to force inputs across decision boundaries. Standard metrics cannot evaluate a model's vulnerability to such deliberate, worst-case input craftings.

2. LABORATORY 02: FGSM EVASION & ROBUSTNESS EVALUATION

2.1 Evasion Attack Methodology

Inference-time evasion testing was conducted using the Fast Gradient Sign Method (FGSM). FGSM computes the sign of the cross-entropy loss gradient with respect to the input features X and applies an L_∞-bounded perturbation:

$$X_{adv} = \text{np.clip}(X + \epsilon \cdot \text{sign}(\nabla_X L(\theta, X, y)), 0.0, 1.0)$$

$$X_{adv} = \text{np.clip}(X + \epsilon \cdot \text{sign}(\nabla_x L(\theta, X, y)), 0.0, 1.0)$$

2.2 Robustness Performance Curve

Epsilon (ϵ)	Attacked Test Accuracy	Performance Delta vs. Clean Basel
0.00	96.22%	Reference Baseline
0.03	90.67%	-5.55%
0.07	79.11%	-17.11%
0.10	59.33%	-36.89%
0.12	51.56%	-44.66%
0.18	20.44%	-75.78%

2.3 Perturbation Analysis & Visual Impact

- Prediction Vulnerability: At $\epsilon=0.12$, true digit class 3 is misclassified as digit 8 while maintaining an imperceptible visual shift to standard human evaluation.
- Perturbation Bound Check: The maximum observed feature displacement was verified at $\max(|X_{adv}-X|)=0.12$, matching the defined L_∞ constraint and validating feature clipping.

2.4 Experimental Limitations

1. Low-Resolution Feature Space: Results are restricted to 8×8 images (64 features). High-dimensional camera inputs exhibit larger adversarial search spaces.
2. Architecture Constraints: Testing was performed exclusively on a linear multinomial logistic regression classifier; evasion dynamics differ on deep neural network architectures.
3. Threat Model Assumptions: Full white-box gradient access overstates attacker capabilities compared to real-world rate-limited API scenarios.

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3. LABORATORY 03: TRAINING DATA POISONING & DATA INTEGRITY

3.1 Label Corruption Simulation

Training set corruption was evaluated by randomly altering a defined fraction of training set labels y_{train} while leaving the test dataset y_{test} pristine. Changed labels were assigned via modular offset: $y_{poisoned}=(y_{original}+\Delta)(\mathrm{mod}10)$.

3.2 Corruption Impact Results

Corruption Rate (%)	Poisoned Training Labels (n*n*)	Clean Test Accuracy	Operational Impact
0%	0	96.22%	Uncorrupted Baseline
5%	67	96.00%	Minimal noise tolerance
10%	135	95.78%	Slight degradation
20%	269	93.33%	Measurable Accuracy loss

3.3 Data Integrity Control Register

Control Domain	Specific Technical / Procedural Implementation
Dataset Provenance	Maintain cryptographic registers tracking data origins and ingestion pipelines.
Cryptographic Hashing	Compute SHA-256 digests for raw datasets; flag any unexpected hash modifications.

Control Domain	Specific Technical / Procedural Implementation
Access Control (RBAC)	Restrict write access to raw training data storage using role-based permissions.
Versioned Labels	Audit label modifications with mandatory reviewer signature logs prior to training.
Anomaly Detection	Run noise-detection algorithms and distance checks to identify mislabeled samples.
Staged Ingestion	Enforce manual approval gates before updating production model training sets.

4. LABORATORY 04: DEFENSIVE RETRAINING & ROBUSTNESS ASSESSMENT

4.1 Defensive Augmentation Configuration

A defensive model variant was constructed by augmenting the original training set X_{train} with an equal number of FGSM-perturbed training examples generated at $\epsilon = 0.10$. The combined training set ($n = 2{,}694$) was used to fit a new logistic regression classifier.

4.2 Comparative Evaluation Results

Model Variant	Clean Accuracy	FGSM Accuracy (ϵ=0.12ϵ=0.12)	PGD Accuracy (ϵ=0.12ϵ=0.12, 7 steps)
Baseline Model	96.22%	51.56%	48.89%
Augmented Defense Model	94.00%	28.44%	26.00%

4.3 Technical Analysis of Negative Results

The augmented defense model exhibited a degradation in both clean performance and adversarial robustness against adaptive attacks:

- Clean Accuracy Drop: Decreased from 96.22% to 94.00%.
- Evasion Robustness Loss: Evasion accuracy under adaptive FGSM dropped from 51.56% to 28.44%, while PGD accuracy dropped from 48.89% to 26.00%.

Cause: Naive single-step FGSM augmentation on linear models can distort decision boundaries without expanding the model's capacity to represent non-linear decision bounds. When an adaptive attack calculates gradients directly against the newly defended model, it easily finds new evasion vectors. This confirms that defenses must be evaluated adaptively against the target model itself rather than assuming transferability.

5. ROBUSTNESS DECISION MEMO & CONTROL RECOMMENDATIONS

5.1 Formal Decision Memo

- Subject: Production Readiness Assessment for Adversarially Retrained Digit Classifier
- Status: REJECTED for Deployment

- Rationale: The candidate defense variant failed to improve system robustness under adaptive FGSM and PGD evaluation, suffering a significant drop in robustness compared to the baseline model.

5.2 Recommended Defensive Safeguards

1. Input Pre-Processing & Gradient Masking Reduction: Implement non-linear smoothing or spatial squeeze filtering on input vectors prior to model inference.
2. Multi-Step Adversarial Training (PGD-based): Retrain models using PGD-generated adversarial inputs across varying perturbation budgets rather than relying on single-step FGSM.
3. Out-of-Distribution (OOD) Detection: Deploy input density estimators to reject samples featuring high gradient magnitude or anomalous noise signatures.
4. Rate-Limiting & Query Obfuscation: Restrict query volume per user and apply confidence-score rounding to limit white-box gradient reconstruction by external adversaries.

APPENDIX: REPRODUCIBILITY MANIFEST & EVIDENCE INDEX

A.1 Environment Specification

- Operating System: Ubuntu 22.04 LTS via WSL2 on Windows 11
- Execution Environment: Python 3.10+, Jupyter Notebook 7.x
- Core Libraries: numpy (1.24+), scikit-learn (1.2+), matplotlib (3.7+)
- Random Seed Policy: SEED = 42 for all train/test splits and model initialization.

A.2 Evidence Index

Evidence ID	Description	Source Notebook	Status
E-01	Baseline clean accuracy (96.22%) and confusion matrix	Lab01_Baseline	Verified
E-02	FGSM robustness curve (ϵ=0.00ϵ=0.00 to 0.180.18)	Lab02_FGSM_Evasion	Verified
E-03	Label corruption test results (0% to 20% corruption)	Lab03_Poisoning	Verified
E-04	Comparative defense evaluation table (Clean/FGSM/PGD)	Lab04_Defences	Verified
E-05	Signed Data Integrity Control Register & Decision Memo	Final Report	Completed